Master Manuscript V4: Structural Consistency and Residue Sanity

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August 16, 2026

1 The Residue Sanity Gate: Metric-Volumetric Invariance

To ensure structural consistency in the EABC framework, we establish the Residue Sanity Gate. For any full-dimensional Euclidean lattice $\Lambda \subset \mathbb{R}^d$, the radial Epstein-Zeta function is defined as:

$$Z_\Lambda^{\text{len}}(s) = \sum_{0 \neq x \in \Lambda} \|x\|^{-s}.$$

The fundamental geometric consistency condition at the pole $s = d$ is:

$$\boxed{\operatorname*{Res}_{s=d} Z_\Lambda^{\text{len}}(s) = \frac{2\pi^{d/2}}{\Gamma(d/2)\,\operatorname{covol}(\Lambda)}} \tag{1}$$

Crucial Clarification on Invariance: The residue is *not* a topological invariant, but a *metric-volumetric lattice invariant*. Under dilation $\Lambda \mapsto c\Lambda$, the zeta function transforms as $Z_{c\Lambda}^{\text{len}}(s) = c^{-s} Z_\Lambda^{\text{len}}(s)$, yielding:

$$\operatorname*{Res}_{s=d} Z_{c\Lambda}^{\text{len}} = c^{-d} \operatorname*{Res}_{s=d} Z_\Lambda^{\text{len}}.$$

More generally, for any $A \in GL(d, \mathbb{R})$, we have $\operatorname{covol}(A\Lambda) = |\det A| \operatorname{covol}(\Lambda)$, leading to:

$$\boxed{\operatorname*{Res}_{s=d} Z_{A\Lambda}^{\text{len}} = |\det A|^{-1} \operatorname*{Res}_{s=d} Z_\Lambda^{\text{len}}.}$$

Thus, the residue is invariant under all volume-preserving linear transformations ($|\det A| = 1$), including orientation-reversing ones.

2 Divisor Information and Winding Audit (Argument Principle)

For a meromorphic function, the winding number along a closed contour γ in the resolvent set is given by the argument principle:

$$\nu_\gamma = \frac{1}{2\pi i} \oint_\gamma \frac{Z'(s)}{Z(s)}\, ds = N_Z(\gamma) - N_P(\gamma) \in \mathbb{Z}.$$

This integer is invariant under homotopies of the contour within the complement of the divisor. This forms the clean dichotomy:

$$\begin{aligned} \operatorname{Res}_{s=d} Z_\Lambda &\quad\longleftrightarrow\quad \text{metric/volumetric pole strength,} \\ \frac{1}{2\pi i} \oint_\gamma d\log Z_\Lambda &\quad\longleftrightarrow\quad \text{discrete divisor / winding information.} \end{aligned}$$

3 Comparative Divisor and Winding Audit for D_4 and E_8

Because D_4 lives in dimension 4 and E_8 in dimension 8, there is no canonical continuous cross-dimensional transition along which a single winding number could be transported directly from D_4 to E_8. Instead, we perform a rigorous comparative divisor audit by analyzing the divisors $\operatorname{div} Z^{\text{len}}_{D_4}$ and $\operatorname{div} Z^{\text{len}}_{E_8}$ using concentric contours centered at the respective dimensional poles:

$$\gamma_{D_4}(R) : |s-4| = R, \quad \gamma_{E_8}(R) : |s-8| = R.$$

The audit maps the discrete step functions:

$$R \mapsto \big(\nu_{D_4}(R), \nu_{E_8}(R)\big),$$

designed to determine the radial distances at which zeros and secondary poles appear and to identify cancellations among factor singularities. No cross-dimensional transport law is assumed.

4 Monomial Fourier–Projection Uniqueness Theorem

Define the $L = 1$ projection functional

$$\mathcal{P}[f] := 2\int_0^1 (1-s) f(s)\, ds$$

and, for integers $a \geq 0$ and $m \geq 1$, the monomial Fourier family

$$f_{a,m}(s) := s^a \sin(2\pi m s).$$

Set $k = 2\pi m$ and

$$M_a(k) := \int_0^1 s^a \sin(ks)\, ds.$$

For $a \geq 2$, two integrations by parts, together with $\sin k = 0$ and $\cos k = 1$, give

$$M_a(k) = -\frac{1}{k} - \frac{a(a-1)}{k^2} M_{a-2}(k), \tag{2}$$

with initial values $M_0(k) = 0$ and $M_1(k) = -1/k$.

Introduce

$$x := \frac{1}{k^2}, \qquad J_a(x) := kM_a(k).$$

Then

$$J_0 = 0, \qquad J_1 = -1,$$

and

$$J_a(x) = -1 - a(a-1)xJ_{a-2}(x), \qquad a \geq 2. \tag{3}$$

Hence $J_a \in \mathbb{Z}[x] \subset \mathbb{Q}[x]$ for every a.

Since

$$\mathcal{P}[f_{a,m}] = 2(M_a - M_{a+1}),$$

define

$$D_a(x) := J_a(x) - J_{a+1}(x).$$

Then

$$\mathcal{P}[f_{a,m}] = \frac{2}{k} D_a\left(\frac{1}{4\pi^2 m^2}\right). \tag{4}$$

Theorem 4.1 (Monomial Fourier–Projection Uniqueness)**.** *For every $a \in \mathbb{N}_0$ and $m \in \mathbb{N}$,*

$$\boxed{\mathcal{P}[s^a \sin(2\pi m s)] = 0 \iff a = 1.}$$

Proof. For $a = 1$, the recursion gives $J_1 = J_2 = -1$, hence $D_1 \equiv 0$. For $a = 0$, $D_0 = 1$. For $a = 2$, one finds $D_2(x) = -6x \not\equiv 0$. For every $a \geq 3$, the recursion yields

$$J_a(x) = -1 + a(a-1)x + O(x^2), \qquad J_{a+1}(x) = -1 + a(a+1)x + O(x^2),$$

so

$$D_a(x) = -2ax + O(x^2),$$

and therefore D_a is a nonzero polynomial in $\mathbb{Q}[x]$.

For fixed $m \geq 1$,

$$x_m = \frac{1}{4\pi^2 m^2}$$

is transcendental because π is transcendental. A nonzero polynomial with rational coefficients has only algebraic roots. Consequently $D_a(x_m) = 0$ can occur only when $D_a \equiv 0$, which happens exactly for $a = 1$. □

The cancellation at $a = 1$ is nontrivial at the level of moments:

$$M_1(k) = M_2(k) = -\frac{1}{k} = -\frac{1}{2\pi m} \neq 0,$$

while

$$\mathcal{P}[s\sin(2\pi m s)] = 2(M_1 - M_2) = 0.$$

Thus the linear exponent is the unique kernel exponent within this monomial Fourier family. This statement is algebraic-analytic; no physical vacuum interpretation is implied by the theorem itself.

5 Residue Specializations for E_8 and D_4

For the standard E_8 lattice, $\mathrm{covol}(E_8) = 1$, and therefore

$$\boxed{\operatorname*{Res}_{s=8} Z^{\mathrm{len}}_{E_8}(s) = \frac{\pi^4}{3}.}$$

For the standard root lattice

$$D_4 = \{x \in \mathbb{Z}^4 : x_1 + x_2 + x_3 + x_4 \equiv 0 \pmod 2\},$$

we have $\mathrm{covol}(D_4) = 2$, hence

$$\boxed{\operatorname*{Res}_{s=4} Z^{\mathrm{len}}_{D_4}(s) = \pi^2.}$$

If D_4 is rescaled to covolume 1, the corresponding residue is $2\pi^2$. These values are normalization-sensitive consequences of the general residue formula and constitute mandatory consistency checks for any proposed product or Dirichlet-series representation.

5.1 Status of the comparative winding stairs

The comparative winding functions are defined by

$$\nu_{D_4}(R) = \frac{1}{2\pi i}\oint_{|s-4|=R} d\log Z^{\mathrm{len}}_{D_4}(s), \qquad \nu_{E_8}(R) = \frac{1}{2\pi i}\oint_{|s-8|=R} d\log Z^{\mathrm{len}}_{E_8}(s).$$

Their step locations are determined by the complete divisors of the corresponding meromorphic continuations. A proposed factorization may be used for this purpose only after it has passed both the pole-structure check and the Residue Sanity Gate. Accordingly, the comparative staircase is defined here as an audit object; a full D_4 staircase is not claimed until an independently validated D_4 factorization or divisor description is supplied.

6 Claim-Wall

- **Metric-Volumetric Residue Formula:** PROVEN
- E_8 **specialization** (covol $= 1$) **and standard** D_4 **specialization** (covol $= 2$)**:** PROVEN
- **Argument Principle / Winding Invariance:** PROVEN
- **Konkrete Produktdarstellung:** VALIDATED (only if pole structure matches the Gate)
- **Comparative Divisor Audit (**D_4 **vs** E_8**):** DEFINED / AUDIT PENDING
- **Monomial Fourier–Projection Uniqueness Theorem:** PROVEN
- **Physical / anyonic interpretation of divisor or winding data:** HEURISTIC / NOT CLAIMED